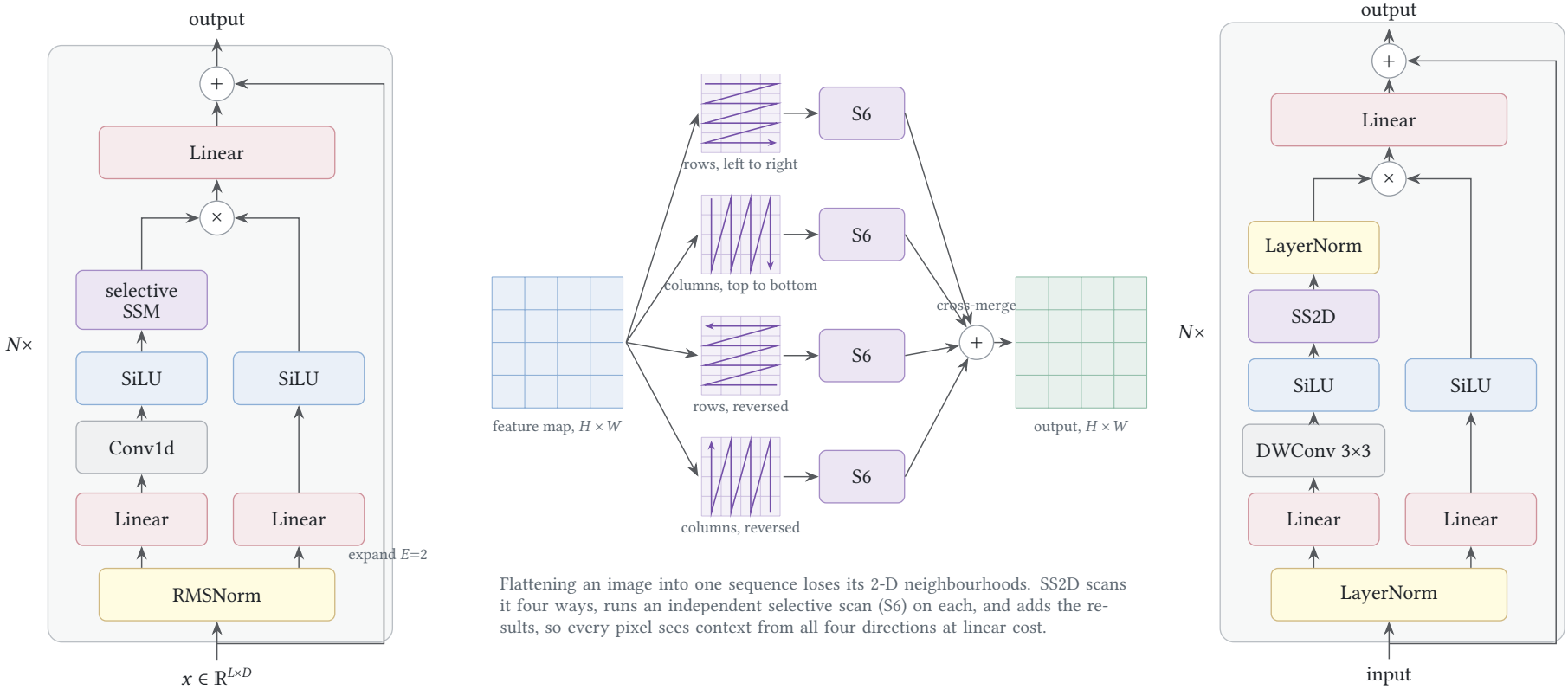
(a) Mamba block
output
Linear
selective SSM
SiLU
SiLU
Conv1d
Linear
Linear
RMSNorm
N×
expand E=2
x ∈ ℝ^{L×D}
(b) 2-D selective scan (SS2D)
rows, left to right
columns, top to bottom
rows, reversed
columns, reversed
S6
S6
S6
S6
feature map, H × W
cross-merge
output, H × W
Flattening an image into one sequence loses its 2-D neighbourhoods. SS2D scans it four ways, runs an independent selective scan (S6) on each, and adds the results, so every pixel sees context from all four directions at linear cost.
(c) VSS block
output
Linear
LayerNorm
SS2D
SiLU
SiLU
DWConv 3×3
Linear
Linear
LayerNorm
N×
input